

ORF309/MAT380

**Basic principles: Foundations
(pp.5-12)**

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Agenda for today

- Basic principles that help us model randomness
 - **Random experiments,**
 - **Sample space,**
 - **Events**
 - **Probability measure**
 - Examples & Exercises
- Mathematical representation
of common sense!

Agenda for today

Random Experiments

Experiments

- **Fact:** Randomness exists in every aspect of life
- **Fact:** We perform experiments all the time!
- **Experiment:** it is any procedure or process that...
 - can be repeated under the same conditions
 - has a well-defined set of possible outcomes
 - produces one outcome per repetition (trial)
- Some experiments are **Deterministic** and some other are **Random**

Deterministic Experiments

- **Deterministic experiment (DE):** it produces the same outcome at every trial, and under the same conditions
 - Its outcome is known or predictable with certainty (and perhaps it is boring to observe)
 - **Examples:**
 1. **Drop a ball in a vacuum** (no air resistance + laws of Physics = 0 randomness)

Deterministic Experiments

- **Deterministic experiment (DE):** it produces the same outcome at every trial, and under the same conditions
 - Its outcome is **known** or **predictable with certainty** (and perhaps it is **boring** to observe)
 - **Examples:**
 1. **Drop a ball in a vacuum** (no air resistance + laws of Physics = 0 randomness)
 2. **Solve a quadratic equation:** $x^2 - 6x + 9 = 0$ (same coefficients give same solution (any time, any place...))

Random Experiments

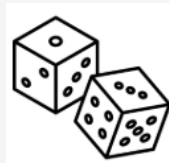
- **Random experiment (RE):** its outcome cannot be predicted with certainty before the trial, and under the same conditions

- **Examples:**

1. Tossing a coin



2. Rolling two dice



3. Observing the stock price of Nvidia today



4. Counting the number of customers entering a store in 1 hour



Sample space

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- The **sample space** (Ω) of a RE is the set of all the **possible outcomes** of the RE
- **Example 1:** RE: throw 2 dice:

$\Omega =$

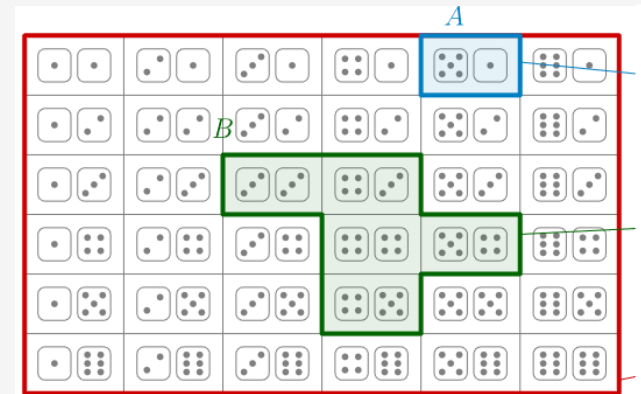
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- Example 1:** RE: throw 2 dice:

$$\Omega = \{(i, j) : 1 \leq i, j \leq 6, i, j \in N\}, \quad |\Omega| = 6^2 = 36$$

$A = (5, 1) \in \Omega \rightarrow$ 1st die shows **5** and 2nd die shows **1**



Wikipedia

Finite set of countable elements

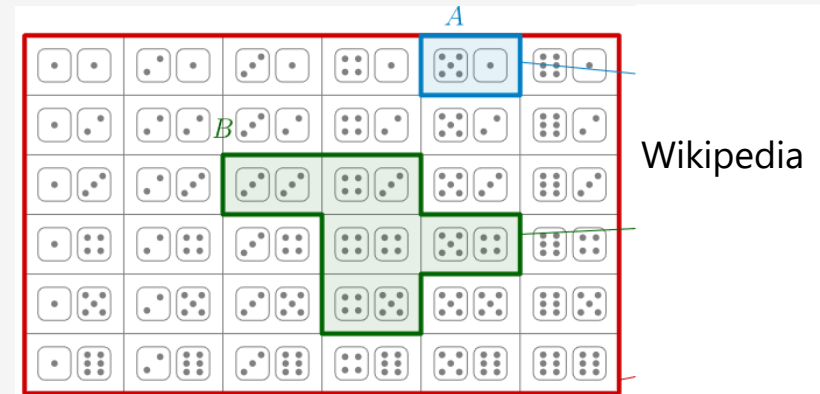
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- Example 2:** RE: How long does it take until a bus arrives at the bus stop?

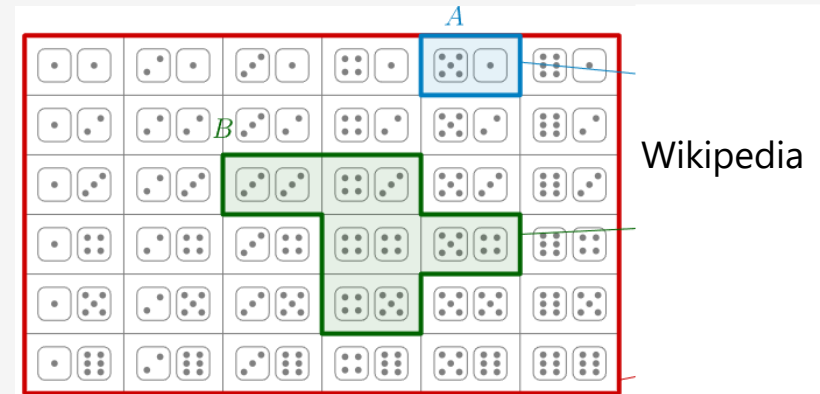
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- Example 2:** RE: How long does it take until a bus arrives at the bus stop?

$\Omega = [0, \infty) \rightarrow$ the bus arrives at some time in the future

$\Omega = [0, \infty) \cup \{\infty\} \rightarrow$ the bust may never arrive

Infinite set of uncountable elements

Sample space

- **Example 3:** RE: You just bought your first car and experiment with it, by measuring its velocity for the first 5 seconds
- Outcomes:

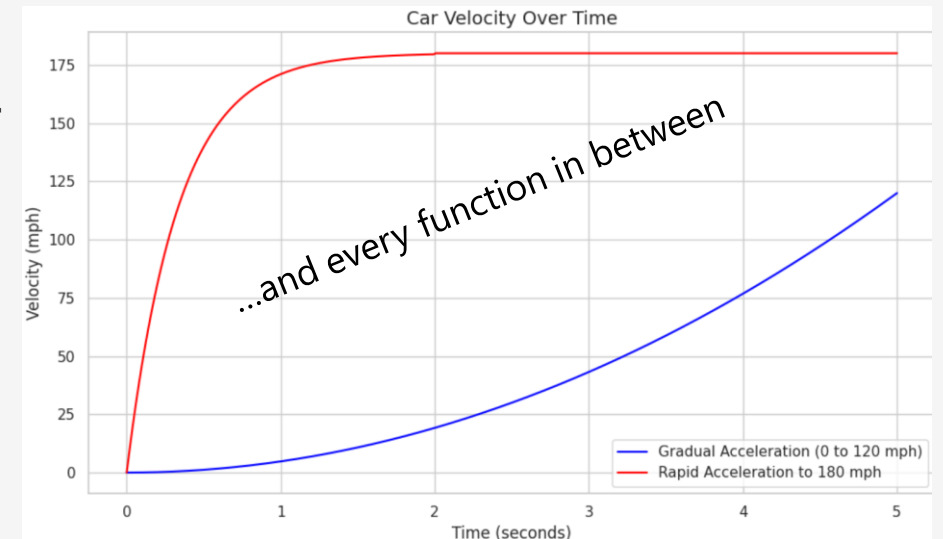
Sample space

- **Example 3:** RE: You just bought your first car and experiment with it, by measuring its velocity for the first 5 seconds
- Outcomes: change of velocity as a **function** of time

$$\Omega = \{v: [0,5] \rightarrow [0, \infty)\}$$

$$\Omega = \{v: [0,5] \rightarrow [0, 211 \text{ miles/h}]\}$$

Maximum speed



Set of all continuous and twice differentiable functions

Events

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- **Example 2:** RE: Throw 2 dice $\rightarrow$ Possible Event: $B = \text{"the sum of the outcome is 7"}$

It is **YES** for the outcomes: $\{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\} \subseteq \Omega$

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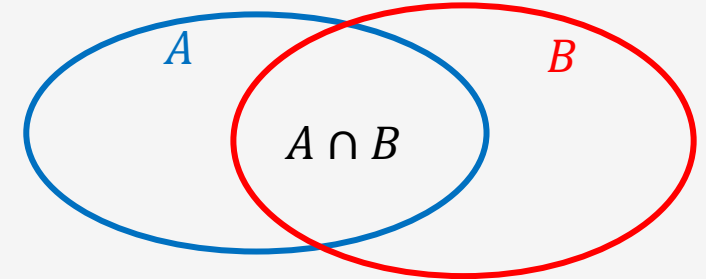
It is **YES** for the outcomes: $\{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\} \subseteq \Omega$

Observation: we can think an event as the **subset** of Ω that makes the statement **TRUE**

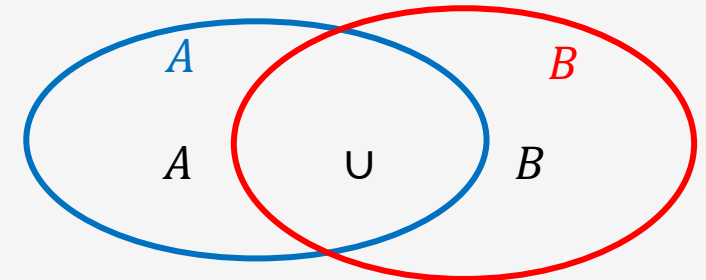
Formal Definition: An event is a **subset** of the sample space Ω

Events – Operations (Set Theory is your best friend!)

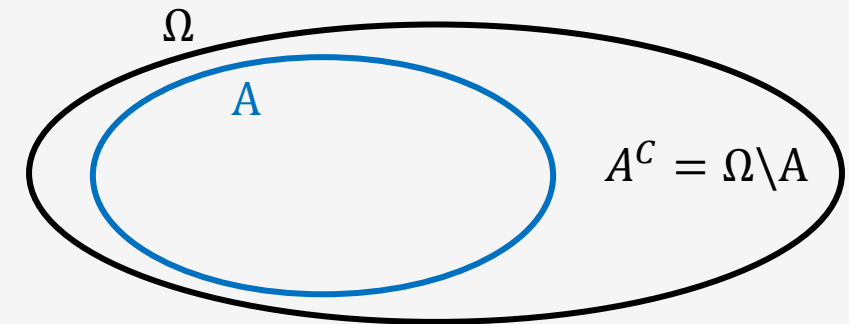
- Event A happens **AND** event B happens: $A \cap B$



- Event A happens **OR** event B happens: $A \cup B$



- An event **does not happen**: $A^C = \Omega \setminus A$



$A^C \rightarrow$ the **complement** event of A

$$\Omega = A \cup A^C$$

Agenda for today

Probability measure (or simply, Probability)

Probability measure

- **Fact:** an event A happens or does not happen in a RE
- We can assign the **degree of confidence** we have that A will happen
 - This is a **number** $\rightarrow P(A)$
- If $P(A) = 1$ we are 100% confident that event A will happen (always happens)
- If $P(A) = 0$ we are 0% confident that event A will happen (never happens)
- If $P(A) = 0.7$ we are 70% confident event A will happen and 30% it will not

Probability measure

- **Definition:** **Probability measure** or just **Probability**, is a rule that assigns a number $P(A)$ to each event A of the sample space Ω such that the following **axioms** hold:
 1. $0 \leq P(A) \leq 1$ ($P(A)$ is a degree of confidence)
 2. $P(\Omega) = 1$ (something will always happen with certainty)
 3. If A, B are two mutually exclusive events ($A \cap B = \emptyset$) then
$$P(A \cup B) = P(A) + P(B)$$
 (additivity property for mutually exclusive events)

Kolmogorov - 1933

Probability measure

- **Generalization** of the additivity property of mutually exclusive events
 - If we have a sequence of events $A_1, A_2, \dots, A_n$ such that

$$A_i \cap A_j = \emptyset, \quad \forall i \neq j$$

then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Simple derivations from the 3 Axioms

- What is $P(A^c) = ?$

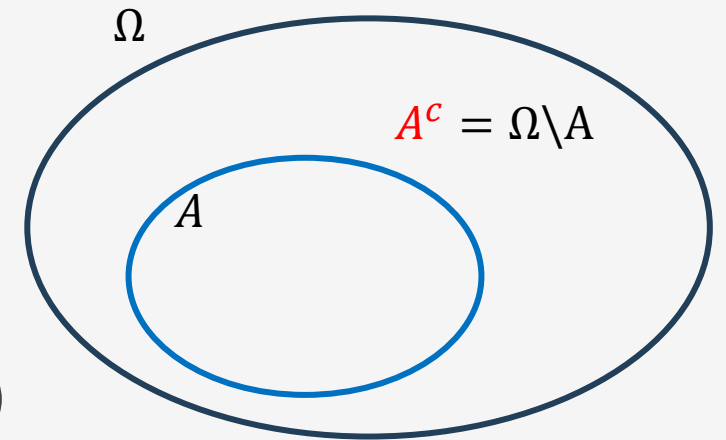
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- **CLAIM:** $P(A^c) = 1 - P(A)$, (we'll prove it!)

Simple derivations from the 3 Axioms

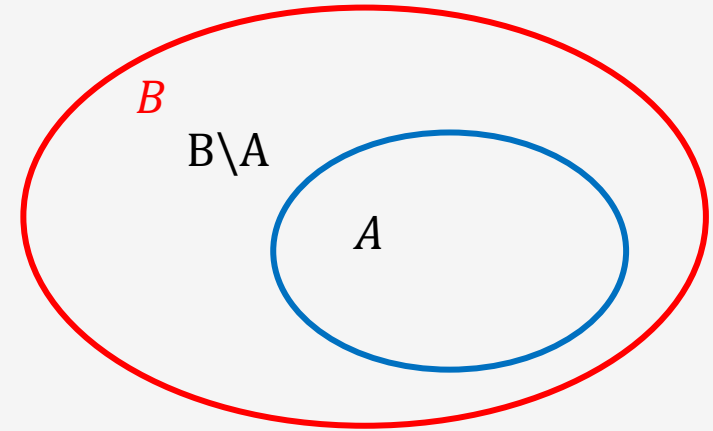
- What is $P(A^c)$ =?
- **CLAIM:** $P(A^c) = 1 - P(A)$
 - We can write: $\Omega = A \cup A^c$
 - We can also note: $A \cap A^c = \emptyset$ (mutually exclusive)
 - Taking probabilities in both sides:

$$P(\Omega) = P(A \cup A^c) \Rightarrow 1 = P(A) + P(A^c) \Rightarrow P(A^c) = 1 - P(A)$$



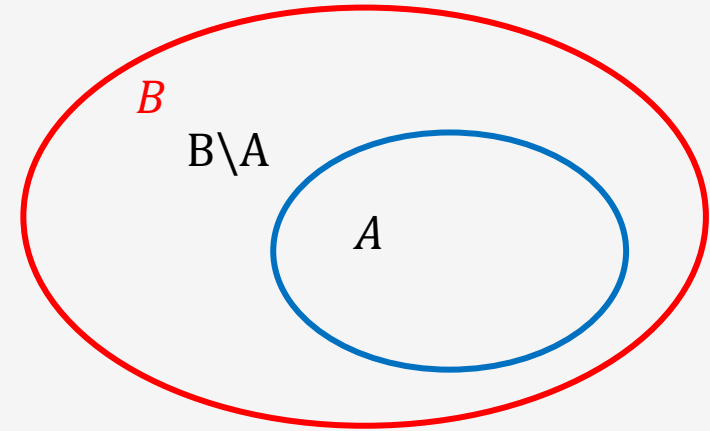
Simple derivations from the 3 Axioms

- Assume you have 2 events A, B
- Assume also that A **implies** B , i.e., $A \subseteq B$
- **CLAIM:** $P(A) \leq P(B)$



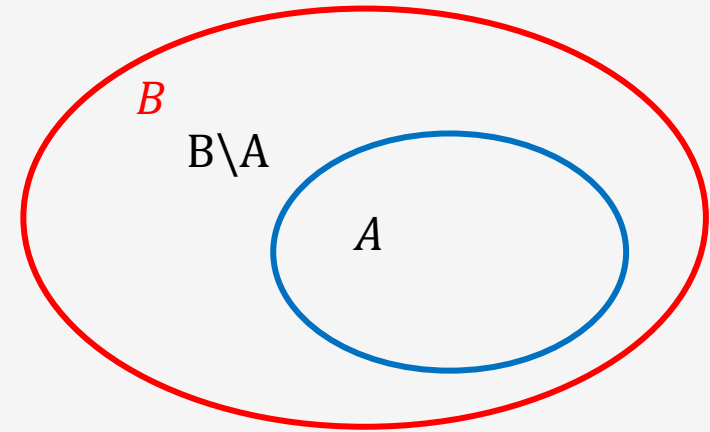
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Simple derivations from the 3 Axioms

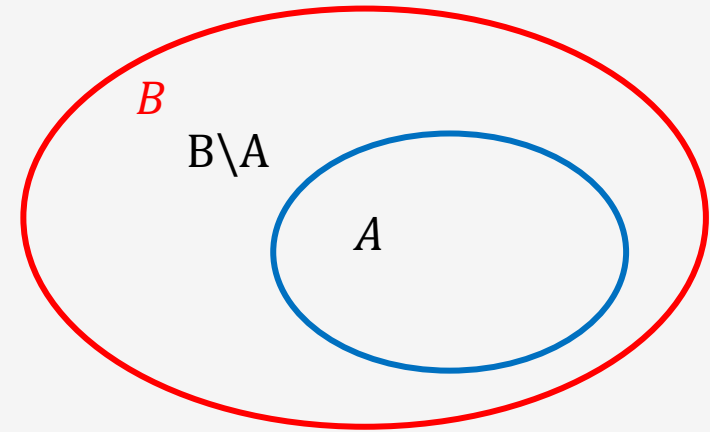
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 - Write: $B = A \cup (B \setminus A)$ and note that $A \cap (B \setminus A) = \emptyset$
 - Take probabilities: $P(B) = P(A \cup (B \setminus A)) = \underset{\geq 0}{P(A)} + \underset{\geq 0}{P(B \setminus A)} \geq P(A)$



Example:

Simple derivations from the 3 Axioms

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Example:

A = "it is raining", B = "It is cloudy". $A \subseteq B \Rightarrow P(A) \leq P(B)$

Exercise 1

- Recall Kolmogorov's Axioms:
 1. $0 \leq P(A) \leq 1$ (degree of confidence)
 2. $P(\Omega) = 1$ (something must happen)
 3. IF $A_i \cap A_j = \emptyset, \forall i \neq j$ then $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$ (prob. of mutually exclusive events)
- Show that the requirement $P(A) \leq 1$ is redundant
- That is, you can derive that $P(A) \leq 1$ for every event A from these axioms, and hence you have the 3 equivalent axioms:
 1. $0 \leq P(A)$ (degree of confidence)
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Exercise 2

- Assume you have n mutually exclusive events, $A_1, \dots, A_n$ and $A_i \cap A_j = \emptyset, \forall i \neq j$.
- Assume also that $\Omega = A_1 \cup A_2 \cup \dots \cup A_n$
- Let B be any event.
- Show that

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + \dots + P(A_n \cap B)$$

Exercise 3

- Show that for any events A and B the following always holds:

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

